



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.NOS.4

Find Factors and Multiples

Lesson 6-7 • Teacher Copy

Name: _____

Date: _____

Class: _____

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can find the greatest common factor (GCF) of two numbers by listing or comparing their factors.

Language Objective: I can explain how I found the GCF using the words factor, common factor, divisible, and greatest common factor.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Greatest Common Factor

Factor

Common factor

Divisible

Prime factorization



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

— The greatest whole number that divides two or more numbers with no remainder.

2

A number that divides evenly into another number with no remainder is a

.

3

A factor that two or more numbers share is a

factor.

4

A number that can be divided by another with no remainder is

by that number.

5

— Writing a number as prime numbers multiplied together. It helps you find the GCF.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How does the GCF help the baker? How many boxes can she make?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Greatest Common Factor** — The greatest whole number that divides two or more numbers with no remainder.
Máximo común divisor — El mayor número entero que divide dos o más números sin dejar residuo.

- ☐ **Factor** — A factor of a whole number divides it evenly, with no remainder. For example, 3 is a factor of 12 because $12 \div 3 = 4$.

Factor — Un factor de un número entero lo divide exactamente, sin dejar residuo. Por ejemplo, 3 es un factor de 12 porque $12 \div 3 = 4$.

- ☐ **Common factor** — A number that divides two or more numbers evenly.
Factor común — Un número que divide a dos o más números sin que sobre nada.

- ☐ **Divisible** — A number you can divide evenly, with nothing left over.

Divisible — Un número que se puede dividir sin que sobre nada.

- ☐ **Prime factorization** — Writing a number as prime numbers multiplied together. It helps you find the GCF.
Factorización prima — Escribir un número como números primos multiplicados. Ayuda a encontrar el MCD.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The baker can make ____ boxes because the GCF of 48 and 32 is ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The number of pods must divide BOTH 24 and 36 evenly.

Evidence: Students see the number of pods has to be a common factor of 24 and 36, so it must divide both with nothing left over, ruling out numbers that only divide one of them.

Reasoning: This evidence connects the definition of factor to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The baker can make ____ boxes because the GCF of 48 and 32 is ____.
- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
- Mi afirmación es ____.*
- I know because ____.

Lo sé porque ____.

- So ____.

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**

Mi afirmación es ____.

- **My math evidence is ____.**

Mi evidencia matemática es ____.

- **This proves ____ because ____.**

Esto demuestra ____ porque ____.

4. Check Your Explanation

☐

I answered the question.

Respondí la pregunta.

☐

I used math evidence: numbers, an equation, a model, or a comparison.

Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

☐

I explained how the evidence proves my answer.

Explicué cómo la evidencia demuestra mi respuesta.

☐

I used at least two math words.

Usé por lo menos dos palabras matemáticas.

☐

I reread complete sentences and punctuation.

Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Greatest Common Factor
2. **Notes 2:** Factor
3. **Notes 3:** Common factor
4. **Notes 4:** Divisible
5. **Notes 5:** Prime factorization
6. **Watch for this mistake:** *A common mistake in Greatest Common Factor is confusing it with the Least Common Multiple: some students list the MULTIPLES of both numbers and pick the smallest shared one (getting an answer bigger than either number, like 72 for 24 and 36) instead of listing FACTORS and picking the largest one they share (12). Remember: the GCF splits numbers INTO equal groups, so it can never be bigger than the smaller of the two numbers.*
7. **Try It 1:** A. 12 — *Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36. The greatest shared factor is 12, so 12 pods can be filled equally.*
8. **Try It 2:** A. 20 — *Factors of 40: 1, 2, 4, 5, 8, 10, 20, 40. Factors of 60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60. The greatest shared factor is 20, so 20 bins can be filled equally.*